Самостійна робота 4 (№2)

Linux та локальний PyPI

Debian ARM64 у Docker Desktop; sshd працює як ssh.service.

1. Системні журнали

```
$ bash scripts/linux.sh ps -p 1 -o comm=
systemd
$ bash scripts/linux.sh journalctl -n 10 --no-pager -o short-iso
2026-10-28T13:06:58+0000 linux-lab systemd[1]: Started json-server.service - JSON endpoint for the Molotov lab.
2026-10-28T13:06:59+0000 linux-lab python[1783]: JSON server http://0.0.0.0:8085/
2026-10-28T13:07:10+0000 linux-lab heartbeat.sh[1793]: 2026-10-28T13:07:10+00:00 mytest працює
2026-10-28T13:08:10+0000 linux-lab heartbeat.sh[1797]: 2026-10-28T13:08:10+00:00 mytest працює
2026-10-28T13:09:10+0000 linux-lab heartbeat.sh[1801]: 2026-10-28T13:09:10+00:00 mytest працює
2026-10-28T13:10:10+0000 linux-lab heartbeat.sh[1805]: 2026-10-28T13:10:10+00:00 mytest працює
2026-10-28T13:10:14+0000 linux-lab systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-10-28T13:10:14+0000 linux-lab systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-10-28T13:10:14+0000 linux-lab systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2026-10-28T13:11:10+0000 linux-lab heartbeat.sh[1811]: 2026-10-28T13:11:10+00:00 mytest працює
$ wc -l logs.txt
      10 logs.txt
$ 
```

```
$ bash scripts/linux.sh journalctl -u ssh.service --no-pager
Oct 28 00:26:42 linux-lab systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 28 00:26:42 linux-lab sshd[63]: Server listening on 0.0.0.0 port 22.
Oct 28 00:26:42 linux-lab sshd[63]: Server listening on :: port 22.
Oct 28 00:26:42 linux-lab systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
$ wc -l sshd-logs.txt
       4 sshd-logs.txt
$ 
```

2. I/O та диски

```
$ cat reports/evidence/iostat.txt
Linux 6.12.76-linuxkit (linux-lab)      10/28/26        _aarch64_       (8 CPU)


Device             tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
vda              65.00         0.00     32828.00     16384.00          0      32828      16384


Device             tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
vda              63.37         0.00     32491.09     72998.02          0      32816      73728


$ cat reports/evidence/io-summary.txt
vda: tps (observed IOPS) = 63.37; read kB/s = 0.00; write kB/s = 32491.09
$ 
```

```
$ cat reports/evidence/disks.txt
NAME     SIZE TYPE MOUNTPOINTS
vda       40G disk
└─vda1    40G part /etc/hosts
                   /etc/hostname
                   /etc/resolv.conf
vdb    623.3M disk
$ cat reports/evidence/fdisk.txt
Disk /dev/vda: 40 GiB, 42949672960 bytes, 83886080 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8d4cd9a4

Device     Boot Start      End  Sectors Size Id Type
/dev/vda1        2048 83886079 83884032  40G 83 Linux
$ 
```

IOPS - операції/с; пропускна здатність - обсяг даних/с.

Виміряно активність віртуального диска Docker Desktop.

fdisk підтримує GPT/MBR; parted має resizepart та сценарний режим -s. resizepart не змінює файлову систему.

3. systemd-сервіс

```
$ cat linux/mytest.service linux/heartbeat.sh
[Unit]
Description=Minute heartbeat for the DevOps lab
After=systemd-journald.service

[Service]
Type=simple
WorkingDirectory=/opt/mytest
ExecStart=/opt/mytest/heartbeat.sh
Restart=on-failure
RestartSec=5
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
#!/bin/bash
set -euo pipefail
while true; do
    printf '%s mytest працює\n' "$(date --iso-8601=seconds)" | tee -a /tmp/test.log
    sleep 60
done
$ 
```

```
$ bash scripts/linux.sh systemctl status mytest.service --no-pager -n 2
● mytest.service - Minute heartbeat for the DevOps lab
     Loaded: loaded (/etc/systemd/system/mytest.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-28 00:26:42 UTC; 12h ago
   Main PID: 51 (heartbeat.sh)
      Tasks: 2 (limit: 4699)
     Memory: 552.0K
        CPU: 1.838s
     CGroup: /system.slice/mytest.service
             ├─  51 /bin/bash /opt/mytest/heartbeat.sh
             └─1835 sleep 60

Oct 28 13:11:10 linux-lab heartbeat.sh[1811]: 2026-10-28T13:11:10+00:00 mytest працює
Oct 28 13:12:10 linux-lab heartbeat.sh[1833]: 2026-10-28T13:12:10+00:00 mytest працює
$ bash scripts/linux.sh journalctl -u mytest.service -n 2 --no-pager
Oct 28 13:11:10 linux-lab heartbeat.sh[1811]: 2026-10-28T13:11:10+00:00 mytest працює
Oct 28 13:12:10 linux-lab heartbeat.sh[1833]: 2026-10-28T13:12:10+00:00 mytest працює
$ bash scripts/linux.sh tail -n 2 /tmp/test.log
2026-10-28T13:11:10+00:00 mytest працює
2026-10-28T13:12:10+00:00 mytest працює
$ 
```

WorkingDirectory=/opt/mytest задає поточний каталог процесу. Абсолютний /tmp/test.log від нього не залежить.

4. Збірка пакета

```
$ cat pyproject.toml src/study5_hello/__init__.py
[build-system]
requires = ["setuptools>=69", "wheel"]
build-backend = "setuptools.build_meta"

[project]
name = "study5-hello-lab"
version = "0.1.0"
description = "Навчальний Python-пакет для локального devpi"
readme = "README.md"
requires-python = ">=3.10"
authors = [{name = "Student"}]

[tool.setuptools.packages.find]
where = ["src"]
"""Мінімальний пакет для перевірки складання та встановлення."""

__version__ = "0.1.0"


def hello() -> str:
    """Повернути привітання."""
    return "Hello, DevOps!"
$ 
```

```
$ cat scripts/build_package.sh
#!/bin/bash
set -euo pipefail
cd -- "$(dirname -- "$0")/.."
export NO_COLOR=1
python -m build
python -m twine check dist/*
$ python3 scripts/show_evidence.py reports/evidence/build.txt --tail 3
Successfully built study5_hello_lab-0.1.0.tar.gz and study5_hello_lab-0.1.0-py3-none-any.whl
Checking dist/study5_hello_lab-0.1.0-py3-none-any.whl: PASSED
Checking dist/study5_hello_lab-0.1.0.tar.gz: PASSED
$ ls dist/
study5_hello_lab-0.1.0-py3-none-any.whl study5_hello_lab-0.1.0.tar.gz
$ 
```

5. Локальний PyPI

```
$ bash scripts/linux.sh systemctl is-active devpi.service
active
$ cat reports/evidence/upload.txt
Uploading distributions to http://127.0.0.1:3141/student/lab/
Uploading study5_hello_lab-0.1.0-py3-none-any.whl
Uploading study5_hello_lab-0.1.0.tar.gz
$
```

```
$ cat reports/evidence/install.txt
Looking in indexes: http://127.0.0.1:3141/student/lab/+simple/
Collecting study5-hello-lab==0.1.0
  Downloading http://127.0.0.1:3141/student/lab/%2Bf/356/a2889bff835ea/study5_hello_lab-0.1.0-py3-none-
any.whl (2.2 kB)
Installing collected packages: study5-hello-lab
Successfully installed study5-hello-lab-0.1.0
Version: 0.1.0
Hello, DevOps!
$ bash scripts/linux.sh /opt/study5-verify/bin/python -I -m study5_hello
Hello, DevOps!
$
```

6. CHANGELOG та формати

```
$ cat CHANGELOG.md
# Changelog

## [0.1.0] - 2026-10-28

### Added

- Функція `study5_hello.hello()` і запуск `python -m study5_hello`.
- Складання пакета у форматах wheel та sdist через `pyproject.toml`.
- Сценарій Molotov для перевірки HTTP 200 та JSON-відповіді.

### Fixed

- Початкова версія; виправлень ще немає.

### Security

- Змін безпеки у цій версії немає.
$ 
```

Ознака	**.deb**	**.rpm**
Інструменти	dpkg / APT	RPM / DNF
Структура	ar: control.tar, data.tar	Заголовки та cpio
Залежності	control: Depends	SPEC: Requires

CHANGELOG пов'язує версію зі змінами та допомагає перевіряти релізи й відкат.

7. Molotov

```
$ cat loadtest.py scripts/load_test.sh
"""Навантажувальна перевірка JSON-сервера."""

import os

from molotov import scenario

URL = os.environ.get("LOADTEST_URL", "http://127.0.0.1:8085/")


@scenario(weight=100)
async def json_ok(session):
    async with session.get(URL) as response:
        assert response.status == 200, f"HTTP {response.status}"
        body = await response.json()
        assert isinstance(body, dict) and body.get("status") == "ok", body
#!/bin/bash
set -euo pipefail
cd -- "$(dirname -- "$0")/.."
# Two workers, at most 50 runs each; fail the command if any scenario fails.
molotov -c -w 2 -d 3 -r 50 --delay 0.05 -f 1 loadtest.py
$ 
```

```
$ python3 scripts/show_evidence.py reports/evidence/molotov.txt --tail 6
[W:0] Ran 30 scenarios
[W:0] Ran 40 scenarios
[W:1] Ran 40 scenarios
**** Molotov v2.6. Happy breaking! ****
SUCCESSES: 100 | FAILURES: 0
*** Bye ***
$ bash scripts/linux.sh curl -s -i -w '\n' http://127.0.0.1:8085/
HTTP/1.1 200 OK
Server: BaseHTTP/0.6 Python/3.11.2
Date: Wed, 28 Oct 2026 13:13:08 GMT
Content-Type: application/json; charset=utf-8
Content-Length: 16

{"status": "ok"}
$ 
```

8. GitHub

https://github.com/mag871498-sys/independent-5-linux-pypi

```
$ git remote -v
origin  https://github.com/mag871498-sys/independent-5-linux-pypi.git (fetch)
origin  https://github.com/mag871498-sys/independent-5-linux-pypi.git (push)
$ git log --oneline -12
695a129 (HEAD -> main, origin/main) chore: add plain evidence viewer and close temporary file descriptor
b6dd1b7 test: run Molotov against real HTTP JSON endpoint
bf871a7 docs: add release changelog and package format comparisons
0de049f test: install package from devpi into fresh isolated environment
63c84c9 feat: publish built artifacts to devpi through Twine
d3aee73 feat: configure isolated local devpi index
9a3fe98 feat: build version 0.1.0 as wheel and source distribution
c21a1fa test: verify systemd heartbeat at sixty second intervals
ace1a94 docs: record Linux block devices and partition table
e5e5953 feat: measure virtual disk IOPS with iostat
dfcc696 docs: collect system and SSH journal records
c03753a chore: prepare Linux laboratory with real systemd
$ git branch --show-current
main
$ 
```